

Introduction

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For decades, Artificial Intelligence research was often characterized by grand claims and limited delivery, leading to periods known as "AI winters" where funding and interest waned. Researchers were sometimes branded as those who "claim but do not deliver." However, the mid-1990s marked a pivotal turning point, ushering in the first significant successes that began to reshape public and scientific perceptions of AI's capabilities. These breakthroughs not only demonstrated AI's potential but also ignited a profound debate about the very definition of intelligence when applied to machines.

This lecture delves into these crucial moments, exploring the initial triumphs that pulled AI out of its prolonged winter. We will examine how these successes, particularly in theorem proving and chess, challenged existing notions of creativity and intelligence, forcing a re-evaluation of what it means for a machine to "think." Understanding this historical context is vital for appreciating the ongoing evolution of AI, the philosophical questions it raises, and the dynamic nature of its perceived capabilities as technology advances.

Core Concepts

Term	Definition	Significance
EQP	An algorithm developed in 1996 that successfully proved a mathematical theorem previously believed to be true but lacking formal proof.	Represented AI's first major success, demonstrating its capability in complex logical reasoning and problem-solving, a field that began in the 1950s.
Deep Blue	An IBM-developed supercomputer chess program that famously defeated reigning World Chess Champion Garry Kasparov in 1997.	A landmark achievement that showed machines could surpass human experts in a domain long considered a benchmark of human intelligence.
Deep Thought	The predecessor to Deep Blue, an earlier IBM chess computer	Highlights the iterative development process in AI and the significant

Term	Definition	Significance
	that was not yet capable of defeating Garry Kasparov.	advancements made between generations of AI systems.
Mechanical Turk	An 18th-century chess-playing automaton that was later revealed to be operated by a hidden human, and later, Amazon's crowdsourcing platform.	Illustrates the historical fascination with intelligent machines and the early deception surrounding AI, contrasting with genuine AI progress.
AI Winter	Periods in AI history characterized by reduced funding and interest due to unfulfilled promises and limited practical applications.	Provides context for the significance of the 1990s successes, as they helped pull AI out of one such period of stagnation.
Moving Goalpost	The phenomenon where as soon as an AI system successfully accomplishes a task, the task is reclassified as "not true AI" or "just an algorithm."	Explains the persistent challenge in defining AI, as human perception of intelligence shifts once a problem is solved by a machine.
Heuristic Search Minimax Algorithm	A specific algorithmic approach used by systems like Deep Blue, involving searching possible moves and evaluating positions using heuristic functions.	Represents the underlying computational methods that enabled Deep Blue's success, sparking debate on whether such methods constitute "intelligence."
If it works, it's not AI	A famous adage in the AI community reflecting the moving goalpost phenomenon; once a problem is solved by AI, it's often no longer considered AI.	Captures the frustration and philosophical challenge faced by AI researchers as their achievements are often re-categorized as mere algorithms.

Detailed Analysis

1. The First Glimmers of Success: EQP and Public Perception

After a prolonged period of limited success and "AI winters," the field of Artificial Intelligence finally achieved its first significant breakthrough in 1996. This milestone was marked by an algorithm named EQP, which successfully proved a mathematical theorem. This theorem had long been believed to be true by mathematicians but lacked a formal proof. The ability of EQP to provide this proof was a major achievement, especially considering that the pursuit of logic theorems by machines had begun as early as the 1950s.

The public and media reaction to this event was telling. The New York Times, for instance, reported that an "Argonne lab program has come up with a major mathematical proof." While acknowledging the achievement, there was a noticeable reluctance to attribute creativity or intelligence to the machine. The report implied that if a human had achieved the same feat, it would have been lauded as creative, but for a machine, it was merely a program. This initial response highlighted a deep-seated human tendency to differentiate between human and machine capabilities, often denying machines the label of "intelligent being" and instead categorizing them as mere "devices." This reluctance to acknowledge machine creativity or intelligence would become a recurring theme as AI continued to advance.

2. Deep Blue's Triumph and the Chess Benchmark

The year 1997 brought an even more widely publicized AI success: the defeat of reigning World Chess Champion Garry Kasparov by IBM's Deep Blue computer. Chess had historically been considered a quintessential test of human intellect and strategic thinking, making this victory a monumental event. The lecture notes that chess had a long history as a "prototypical intelligence application," dating back to the "Mechanical Turk" in the olden days. The Mechanical Turk was a famous 18th-century chess-playing automaton that amazed audiences but was later revealed to have a human operator hidden inside. This historical context underscores the long-standing human fascination with machines that could mimic intelligent behavior, and the eventual reality of a true AI system achieving this feat.

Deep Blue's journey to victory was not without its challenges. Its predecessor, "Deep Thought," was not yet capable of defeating Kasparov. In 1996, Deep Blue itself competed against Kasparov but was defeated 4-2. This loss spurred IBM engineers to improve the system, leading to a rematch in 1997. In this highly anticipated second encounter, Deep Blue emerged victorious with a score of 3.5 to 2.5, a close but decisive win. Kasparov's reaction to the defeat was profound; he famously stated that he "could feel human-level intelligence across the room." While this

might seem like a compliment, the lecture suggests it could also be interpreted as a "bad compliment" if the machine is perceived as surpassing human capabilities, implying a loss of human uniqueness in that domain.

The victory sparked intense debate about whether Deep Blue truly used "AI" or was merely a "brute force-ish search system." Deep Blue relied on a "heuristic search minimax algorithm with heuristic evaluation functions." This algorithm involved evaluating millions of possible moves and positions to determine the optimal next step. The question arose: was this intelligence, or just sophisticated computation? Today, no human can defeat the best chess machines, and even a single victory against them would be considered a "triumph of human genius," indicating that humans now need "machine-level intelligence" in chess. This evolution highlights that intelligence is not a binary phenomenon but exists in degrees.

3. The "If it works, it's not AI" Phenomenon and the Moving Goalpost

The debate surrounding Deep Blue's intelligence led to a famous adage in the AI community: "If it works, it's not AI." This statement encapsulates the "moving goalpost" nature of defining artificial intelligence. As soon as an AI system successfully solves a problem, many observers, including some researchers, tend to reclassify the achievement as "just an algorithm" or "algorithm X," rather than true intelligence. Drew McDermott, a senior AI researcher at the time, famously responded to this sentiment by drawing an analogy: "saying Deep Blue does not really think about chess, it is not intelligent about chess, is like saying that an airplane does not fly because it does not flap its wings." His point was that while the method might differ from human intelligence, the outcome—achieving the task—is still a form of intelligence.

The lecture illustrates this "moving goalpost" phenomenon with a compelling example: the "intelligent door." One hundred and twenty years ago, a door that automatically opened when someone approached would have been considered science fiction, a magical display of intelligence. Today, we understand it as "just a motion sensor." The moment the underlying mechanism is understood, the "magic" dissipates, and the perceived intelligence vanishes. This leads to a "deep sense of loss" once understanding replaces wonder. As AI researchers achieve one goal, the definition of "intelligence" shifts to the next unsolved problem. What was once considered AI becomes a solved engineering problem, and the "real intelligence" is always the next, currently intractable challenge. This constant redefinition ensures that AI researchers "will always be in the job," as the frontier of "unsolved intelligence" continuously moves.

4. AI Winters and the Shifting Definition of Intelligence

The history of AI has been marked by cycles of enthusiasm and disillusionment, often referred to as "AI winters." These periods occurred when the ambitious claims of AI researchers failed to materialize into practical applications, leading to a withdrawal of funding and public interest. The successes of EQP and Deep Blue in the mid-1990s were crucial in pulling AI out of one such winter, demonstrating tangible progress after years of perceived stagnation.

The lecture emphasizes that the definition of intelligence itself is fluid and constantly evolving, especially in the context of AI. What is considered intelligent today might be seen as a mere algorithm tomorrow, once its underlying principles are understood and implemented. This dynamic nature means that AI is not a fixed target but a constantly moving one. The field grapples with the paradox that as soon as a problem is solved by AI, it often ceases to be considered "AI" in the popular imagination, becoming instead a specialized algorithm or technology. This continuous re-evaluation of what constitutes "intelligence" is a fundamental aspect of AI research, ensuring that the pursuit of truly intelligent machines remains an ongoing and challenging endeavor.

Key Takeaways

- [] The mid-1990s marked the first significant successes in AI, including EQP proving a mathematical theorem in 1996.
- [] Public and media reactions to early AI successes often showed reluctance to attribute creativity or intelligence to machines, viewing them as mere devices.
- [] Deep Blue's victory over Garry Kasparov in 1997 was a landmark event, demonstrating machine superiority in a domain long considered a benchmark of human intelligence.
- [] Chess has a long history as an intelligence application, dating back to the Mechanical Turk, highlighting human fascination with intelligent machines.
- [] The debate around Deep Blue questioned whether its algorithmic approach constituted "true AI" or just sophisticated computation.
- [] The "moving goalpost" phenomenon describes how tasks solved by AI are often reclassified as "not true AI" once the underlying mechanism is understood.
- [] The adage "If it works, it's not AI" reflects the continuous redefinition of intelligence in the field, where "AI" is always the next unsolved problem.

- [] Understanding the mechanism behind a seemingly magical AI achievement often leads to a "deep sense of loss" of wonder, as seen with the automatic door analogy.
- [] AI's history includes "AI winters," periods of reduced interest and funding, from which the 90s successes helped the field emerge.
- [] Intelligence is not a binary concept but exists in degrees, and its definition shifts as machines achieve new capabilities.

Conclusion

The 1990s represented a critical turning point for Artificial Intelligence, moving the field from theoretical promise to tangible, albeit debated, success. The achievements of EQP in theorem proving and Deep Blue in chess not only showcased the growing capabilities of machines but also ignited profound philosophical discussions about the nature of intelligence itself. These events forced a re-evaluation of what it means for a machine to "think" or be "creative," highlighting the human tendency to shift the definition of intelligence once a problem is solved by artificial means.

This "moving goalpost" phenomenon, encapsulated by the phrase "If it works, it's not AI," continues to shape the perception and direction of AI research. As we move forward, understanding this historical context is crucial for appreciating the ongoing challenges in defining and developing AI. The lecture underscores that the pursuit of artificial intelligence is not merely a technical endeavor but also a continuous exploration of what constitutes intelligence, both human and artificial, ensuring that the field remains dynamic, challenging, and perpetually at the frontier of innovation.